

subspaces that are *not aligned to neuron bases* (Mikolov et al. 2013a; Arora et al. 2016); §6.2 defines and visualizes this concept in more detail.

To overcome this disadvantage, one can generalize causal mediators to include arbitrary *non-basis-aligned* subspaces of $\mathbf{h}^\ell$. This allows us to capture more sophisticated causal abstractions encoded in latent space, such as causal nodes corresponding to greater-than relationships (Wu et al. 2023), or equality relationships (Geiger et al. 2024). Common methods for locating these are discussed in §6.2.

The primary advantage of considering an arbitrary subspace as a mediator is its expressivity *and* precision: subspaces often capture distributed abstractions that are not cleanly aligned to specific neurons. However, they are generally more difficult to locate than basis-aligned components or lower-dimensional directions, as we are required to have specific hypotheses as to how a model accomplishes a task and access to minimal pairs that isolate the target concept. Some optimization-based procedure is also usually required. Thus, non-basis-aligned multidimensional subspaces are generally more selective and human-interpretable than basis-aligned subspaces, and can maintain similar sparsity and generality; however, they may sacrifice faithfulness, as the optimized component can in theory learn concepts that were not in the model itself. This combined with the strict data requirements makes them useful primarily for verifying mechanistic hypotheses, but less useful for explaining model behaviors or localization/editing.

Directions. A recent line of work aims to automatically identify specific directions (one-dimensional non-basis-aligned objects) $\mathbf{d}$ that correspond to monosemantic concept representations; see Eq. 2 for a definition of $\mathbf{d}$. Identifying and labeling these monosemantic model abstractions (often called **features**; Bricken et al. 2023; Cunningham et al. 2024; Huang et al. 2024) can reveal units of computation the model uses to solve tasks in a way that is often easier for humans to interpret.²³

There is also initial evidence that these directions may enable fine-grained model control (Panickssery et al. 2024; Marks et al. 2025; Tigges et al. 2023). Past work has found initial signs that basis-aligned directions could be leveraged to edit (Meng et al. 2022) or steer (Subramani, Suresh, and Peters 2022; Turner et al. 2023) model behavior, whereas more recent work has found that steering using non-basis aligned directions is both more effective and more precise (Marks et al. 2025; Arora, Jurafsky, and Potts 2024; Wu et al. 2025). For example, there is work that uses linear probes to understand the effects of a direction on the model behavior (Chen et al. 2024b; Ravfogel et al. 2020; Elazar et al. 2021; Ravfogel et al. 2021; Lasri et al. 2022; Marks and Tegmark 2023), as well as work that uses these directions to steer model behaviors—e.g., by minimizing (Marks et al. 2025; Cunningham et al. 2024) or amplifying (Templeton et al. 2024) directions corresponding to fine-grained concepts, such as typically female names or the Golden Gate Bridge.

Nonetheless, directions have key disadvantages. The search space over possible non-basis-aligned directions is infinite, making it impossible to exhaustively search over them. To discover them, we are generally *required* to modify the computation graph in some way to obtain some discrete search space—for example, by learning coefficients on each neuron, as in Eq. 2, requires learning new parameters. Regardless of the method, optimization algorithms will introduce confounds due to their stochastic

²³ Note that these directions are not necessarily subspaces of activation space: there are often non-linearities used in computing them, even though the vectors in activation space are involved in computing the directions. Therefore, we will refer to any one-dimensional space as a **direction**, but do not require it to be a subspace of activation space.

nature. In short, non-basis-aligned directions have significant advantages in human-interpretability, selectivity, and sparsity over basis-aligned mediator types, and their data requirements are less strict than non-basis-aligned multidimensional subspaces. This makes them a good starting point if one’s goal is to explain model behaviors. Nonetheless, faithfulness is likely to be worse than that of basis-aligned components, given that optimization is required; generality may also suffer because of reconstruction error, and because the mediators are so fine-grained.

5.4 Non-linear Mediators

Non-basis-aligned directions and subspaces can be the most sparse and selective *linear* mediator types. However, recent work has demonstrated that some features in language models can be represented non-linearly. For example, there exist features that are encoded as vector *magnitudes* in any direction (Csordás et al. 2024). Past work has similarly found that many concepts can be more easily extracted using non-linear probes (Liu et al. 2019a), and that non-linear concept erasure techniques tend to outperform strictly linear techniques (Iskander, Radinsky, and Belinkov 2023; Ravfogel et al. 2022). However, in causal and mechanistic interpretability, most work has thus far tended toward using linear representations as units of causal analysis. Thus, there is significant potential in future work for systematically locating non-linearly-represented features—e.g., using group sparse autoencoders (Theodosios and Ba 2023), which could isolate multiple directions simultaneously, and/or probing and clustering techniques to identify multidimensional features (Engels et al. 2025). Non-linear features have not been extensively studied, despite their expressivity; we therefore advocate investigating these mediators in §7.

6. Searching for task-relevant mediators

Once one has selected a task and a type of mediator, how does one identify task-relevant mediators of that type? The answer depends largely on the type of mediator chosen. If a given mediator type is finite in number—as is the case for sets of model components $\mathbf{h}^\ell$ such as neurons, layers, and submodules—one could perform an exhaustive search over all possible mediators, choosing which to keep according to some metric; §6.1 discusses this approach. However, other mediator types, including non-basis-aligned directions and subspaces, are continuous, rendering an exhaustive search impossible. There are two common solutions to the problem of continuous mediator search spaces: (i) employ optimization to search this space, or (ii) narrow the space into an enumerable discrete set. Both are discussed in §6.2. Table 2 (§5) summarizes the kinds of mediators that tend to be paired with particular search methods. We do not discuss runtimes in this section, but see Appendix B for an overview of the relative computational requirements for each mediator/search method combination.

6.1 Exhaustive search

Suppose we are given a neural network with a finite set of candidate mediators $\{Z_i\}_{i=1}^N$, such as the set of all neurons $\{\mathbf{h}_0^0, \dots, \mathbf{h}_d^\ell\}$. One way to identify task-relevant mediators from this set is to assign each mediator Z_i a task-relevancy score $S(Z_i)$ and then select the mediators with the top scores. S is typically the indirect effect (IE; Pearl 2001; Robins

and Greenland 1992), as defined in Eq. 1.²⁴ Computing this generally entails iterating over each mediator Z_i , setting its activation to some counterfactual value (either from a different input where the answer is flipped, or a value that destroys the information in the neuron, such as its mean value), and measuring how much this intervention changes the output. For example, Vig et al. (2020) and Finlayson et al. (2021) perform counterfactual interventions to the activation of each neuron in an LM, measuring how much each intervention changes the probability of correct completions. This metric is based on the notion of counterfactual dependence, where we measure the difference in some output metric m before and after intervening on a given component Z_i .

Exhaustive searches have many advantages: their results are comprehensive, causally efficacious, and relatively conceptually precise if our mediators are fine-grained units like neurons. We are also not required to have a pre-existing causal hypothesis as to how or where in its computations a model performs a task: we may simply observe how interventions to a model component changes the model’s behavior or the probability of some continuation. Because of these advantages, this method is common when we have a finite set of mediators—for example, in neuron-based analyses (Vig et al. 2020; Geiger et al. 2021; Finlayson et al. 2021) or attention-head-based analyses (Vig et al. 2020; Conmy et al. 2023; Syed, Rager, and Conmy 2023).

However, exhaustive searches also have two significant disadvantages. The most obvious is that, in exact form, an exhaustive search requires $O(N)$ forward passes, where N is the number of mediators. This does not scale efficiently as models scale, both because the number of components increases and because the computational cost of inference scales with model size. This may be why exhaustive searches have not often been extended to *sets* of neurons or heads, as this results in a combinatorial explosion in the size of the search space such that the number of required forward passes increases to $O(2^N)$. Searches over sets of components can be approximated using greedy or top-k approaches, as in Vig et al. (2020), but this is not guaranteed to find the best solution to the problem of assigning causal credit to groups of components.

To overcome these challenge, gradient attributions have become common. These can be conceptualized as fast linear approximations to causal mediation analysis. As they are local linear approximations, gradient attributions are technically not causal²⁵ and not always accurate, but they have far better asymptotic runtime than causal mediation analysis. Example methods include attribution patching (Kramár et al. 2024; Syed, Rager, and Conmy 2023), and improved versions thereof inspired by integrated gradients (Sundararajan, Taly, and Yan 2017; Marks et al. 2025; Hanna, Pezzelle, and Belinkov 2024). These techniques usually entail backpropagating from some target metric m ; this is typically the probability of the correct next token or correct label y , or the probability difference between y and some minimally differing incorrect output y' . This yields $\left. \frac{\partial m}{\partial h_i^\ell} \right|_x$ (the gradient of m with respect to the activation of neuron h_i^ℓ given input x), which can be conceptualized as a local estimate of the slope of m with respect to

24 Other causal metrics include the **direct effect**, which measures the direct influence of the input on the output behavior except via the mediator. While more rarely used, it can be a helpful metric in tandem with indirect effects, as in Vig et al. (2020). There is also the **total effect**, which is the impact of changing the input on the model’s output behavior. Note that the total effect does not directly implicate any particular component in model behavior, as it depends only on the input.

25 Gradient attributions provide a scalar value whose magnitude can be interpreted as a local linear approximation of the model output’s sensitivity to the component. This is conceptually related but distinct from causal analysis—e.g., because it is sensitive to confounding, does not capture non-linear effects, and does not directly test a causal model nor counterfactual hypothesis.